

Fix model, TP and measured frequency $q = (m, t, f)$
Same hardware, engine, PP and measurement scope

Mixed probes

Decode + prefill arrivals
Measure TTFT / stalls
Optional matched power table

Prefill samples

Vary prompt length s
Fit time and mean power
 $\hat{t}_P(s), \hat{P}_P(s)$

Decode samples

Vary batch B / context c
Fit time; interpolate power
 $\hat{t}_D(B, c), \hat{P}_D(B, c)$

Freeze parameters / tables; validate with fresh holdout samples
Attach supported domains and errors; holdout data do not enter fitting

Planner: bounded queries $\rightarrow$ load / queue composition $\rightarrow$ candidate costs

Mixed: $(\hat{E}_M(H), \widehat{\text{TTFT}}_M, \widehat{\text{TPOT}}_M)$ Prefill: $(\hat{E}_P, \hat{t}_P)$ Decode: $(\hat{e}_D, \hat{t}_D)$